

# Amritraj Lamba

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Heritage International Xperiental Learning, Gurugram.

## EDUCATION

**AUGUST 2024 - MAY 2026**

**IB Diploma (Grade 11 and Grade 12)**

- Relevant Coursework: HL Math Analysis and Approaches, HL Chemistry, HL Physics, SL Economics, SL French B, SL English A Language and Literature

**JUL 2023 - JUN 2024**

**IGCSE (Grade 10)**

- 10/10 A\*s
- School Topper in Additional Mathematics, School Topper in Chemistry

**Standardized Test Scores:**

- SAT – 1550 (Maths 800, Reading & Writing 750)
- AP – Calculus BC: 5/5
- IELTS – Band 8

## LEADERSHIP

**Founder, Sewage Suraksha (April 2024 - Present)**

- Developed a sewage lid with multi-gas sensors; it is a low-cost retrofit ensuring safety of municipal workers.
- The lid detects harmful gases, including  $H_2S$ ,  $CH_4$ ,  $NH_3$ , &  $CO$ , and alerts authorities to hazardous gas levels.
- Tested on 17 sewers and cleared by the National Disaster Management Authority for adoption.

**Founder, Flood Shield (July 2024 - Present)**

- Flood detection IoT for real-time cloud mapping and emergency coordination; Special Praise for 'practicality' at the Princetonian.
- Adopted by the National Disaster Management Authority.
- Designed and piloted an affordable early warning system for floods that alerts authorities to areas where flooding occurs.
- Adopted by 1000+ houses and commended by the Haryana Government.
- Also received the Crest Gold Award for this invention.

**Leader — Ingenix School STEM Society (October 2024 - October 2025)**

- Revived and expanded society from 20 to 50+ members.

- Mentored two teams for the Cambridge Science Competition & F1 in Schools, both qualified internationals.

#### **Leader — FUNCTION, After School Robotics Club (October 2024 - October 2025)**

- Founded a 20+ member club teaching CAD, programming, and mechanical design outside of competition teams.
- Introduced collaborative engineering workshops (bridge/tower builds, quizzes) to strengthen teamwork and experimentation.
- Built a hands-on learning culture that became an entry pathway for beginners; several members progressed to competitive robotics.

## **RESEARCH**

#### **Automation & Robot Adoption in Delhi-NCR Factories, Author (January 2025 - September 2025)**

- Analysed five factories from fully manual (e.g., Metaval Engineering) to highly automated (e.g., Kadence Automation) to assess efficiency, training needs, and scalability.
- Found robot-assisted systems improved efficiency by up to 20% while reducing fatigue and errors.
- Highlighted the need for retraining, ergonomic design, and phased adoption.
- Developed a roadmap for SME modernisation without worker displacement; validated through Delhi-NCR field data, mentorship from Stanford's Dr. Arvind Karunakaran, and publication in Lex Localis.

#### **Coral Reefs as Natural Barriers to Coastal Erosion in India, Author (March 2024 - August 2025)**

- Used satellite imagery and GIS to show shorelines protected by healthy reefs had less than half the erosion of areas with degraded reefs.
- Provided evidence to integrate reef restoration into climate adaptation strategies for cities like Mumbai and Chennai, complementing engineered defenses.
- Recognition included publication in the International Journal of Environmental Sciences and mentorship from Prof. Umang Soni, with validation in competitive research platforms.

## **SUMMER PROGRAMS**

#### **University of Pennsylvania, Wharton Global Youth Program (June 2025)**

##### **Product Design Academy**

- Collaborated with Professor Taylor Caputo and international peers under the Wharton faculty, learning design thinking, prototyping, and TAM/SAM/SOM analysis.
- Learned to apply structured market-sizing frameworks to test creative engineering ideas against commercial viability.
- Prototypes developed during the program reflected this rigorous approach, and faculty feedback validated the professional grounding in product innovation.

##### **LaunchX, Online Bootcamp (June 2025)**

- Selected for LaunchX; collaborated with peers to ideate, prototype, and pitch ventures.
- Gained hands-on experience in market validation, business modelling, and product design.

##### **Euler's Circle (June - July 2024)**

- "Serious Mathematics for Serious High School Students" under Stanford Prof. Simon Salzedo.
- One of 30 members worldwide.
- Capstone on linear algebra and projective geometry; improved modeling intuition later applied to CFD and robot mechanisms.
- Attended on 50% scholarship.

## INTERNSHIPS

### **Summer Intern, Vardan Envirolabs, Gurugram (December 2024 - February 2025)**

- Worked in an EIA consultancy, performing lab tests on air and water samples and learning to operate professional instruments like spectrophotometers and gas analyzers.
- Gained practical understanding of how data translates into impact-assessment reports that influence project approvals.
- Used the knowledge from this experience to begin working on the sewage-detection lid project.
- Received a certificate and recommendation letter from the lab director, validating both the internship and the project.

## ADVANCED COURSES

### **Art of Problem Solving (AoPS)**

- Completed all introductory & intermediate courses on Algebra, Number Theory, Geometry, and Counting & Probability.

### **World International Mathematics Olympiad Training Course (January 2024)**

- Professional Mathematical Olympiad training provided by Olympiad Champion Education Group at University of Malaya, Kuala Lumpur, Malaysia

## ACADEMIC HONOURS AND AWARDS

### MATHEMATICS

#### USA Math Olympiads (2022–24)

- AMC 12B (2024): Scored 132/150
- School Rank 1, Country Rank 28, Qualified for AIME, Certificate of Distinction, Honour Roll (Top 2.5%)

#### World International Mathematics Olympiad (WIMO), Final, January 2024 (Kuala Lumpur, Malaysia)

- Bronze Award

#### CEMC (University of Waterloo)

- Pascal Contest (9th Grade, 2023): 2nd Place from India region, Certificate of Distinction, Student Honour Roll
- Fryer Contest (9th Grade, 2023): 2nd Place from India region, Certificate of Distinction
- Cayley Contest (10th Grade, 2024): Certificate of Distinction, Student Honour Roll
- Galois Contest (10th Grade, 2024): Certificate of Distinction
- Fermat Contest (11th Grade, 2025): 9th Place from India region, Certificate of Distinction, Student Honour Roll
- Hypatia Contest (11th Grade, 2025): Certificate of Distinction

#### SIMOC (Singapore International Mathematical Olympiad Challenge), Singapore, July 2024

- Maths Olympiad Contest: Silver Award
- Maths Warrior Challenge: Silver Award
- Mind Sports Challenge: Bronze Award

#### IOQM (Indian Olympiad Qualifier in Mathematics) 2024–25

- Certificate of Merit

#### SASMO (Singapore and Asian School Mathematics Olympiad) 2024 (Grade 10)

- Gold Award

#### AMO (American Mathematics Olympiad)

- Gold Award (2023, Grade 10)
- Gold Award (2022, Grade 9)

#### **TIMO (Thailand International Mathematical Olympiad) Final Round 2022–23**

- Gold Award (Secondary 3 Group)

#### **HKIMO (Hong Kong International Mathematical Olympiad)**

- Heat Round 2024: Gold Award (Senior Secondary Group)
- Final Round 2022: Gold Award (Secondary 2 Group)

#### **Stanford Math Tournament (SMT) 2023**

- Honourable Mention in General Test

#### **Australian Mathematics Competition 2024**

- Certificate of Distinction (Senior Division, Grade 11)

#### **International Maths Kangaroo Competition (2023)**

- Silver Medal

#### **International Junior Honor Society (IJHS), Singapore (2023/2024)**

- Certificate of Membership

### **CHEMISTRY**

#### **Avogadro Chemistry Contest (University of Waterloo) (11th Grade, 2025)**

- Certificate of Distinction

### **PHYSICS**

#### **Sir Isaac Newton Physics Contest (University of Waterloo) (11th Grade, 2025)**

- Certificate of Distinction

#### **Future Intelligence Science Olympiad Physics, International Round (2024)**

- Gold Medal

## ROBOTICS

### First Tech Challenge (FTC), February–July 2024

- Role: CAD and Design Head of Team, Game Player
- Achievements: Qualified for Asian Pacific Open Championship (APOC), Sydney, Australia
- Won: Inspire Award (2nd Place) and Think Award (2024), 4th Ranked in Knockout Races out of 32 teams.

### F1 in Schools 2024

- Role: Design Engineer
- Ranked 15th in Nationals
- Regional Digital Marketing Award

## SOCIAL IMPACT PROJECTS

### Teens of God (Regd. Local NGO) (Aug 2022 - Dec 2023)

- Mentor/Tutor teaching basic Math and Science concepts to underprivileged children.
- One of the lead members of the NGO's robotics branch, where we taught concepts of robotics and circuits using donated STEM kits.

### Signal Stars (FTC Outreach Initiative, Thane) (Jan 2024 – Jul 2024)

- Volunteered as a mentor for the Signal Stars team from a rural-area community in Thane.
- Led rural students through design, construction and competition preparation, culminating in the team earning the **Judges' Choice Award** at the FIRST Tech Challenge India Championship 2023-24
- Guided the students to present a strong engineering portfolio, build a modular robot mechanism and collaborate effectively under competition constraints.

## SKILLS

- Technical Skills/Tools: Python, 3D Modelling (CAD), Arduino, Robotics.
- Sports: Diploma & Black Belt in Shotokan Karate.
- Music: Guitar (Trinity Rock & Pop Grade 5).
- Languages: English, Hindi, French, Punjabi.

## MEMBERSHIP

**New York Academy of Sciences – The Junior Academy Member**

- Selected member; engaged for a year with the Academy's STEM community.
- Part of the Solar Power team under Energy Infrastructure.